



IMDG - UN card

UN 3499

UN number 3499		Shipping name CAPACITOR, ELECTRIC DOUBLE LAYER KONDENSATOR, ELEKTRISK DOBBELTLAGS-				
Class 9	Packing group —	Hazard labels 9	Marine pollutant Maybe	EmS F-A, S-I	ADR transport category: 4 ADR tunnel: (E)	
Hazard labels 		Marking UN 3499 CAPACITOR, ELECTRIC DOUBLE LAYER				
Packaging		Can be packed in suitable packaging according to packing instruction P003.				
Mixed packaging		The substance may be packaged with other substances in the same outer packaging if: - it is permitted to load the substances together, and - the substances do not react dangerously with one another.				
Remarks		This entry applies to electric double layer capacitors with an energy storage capacity greater than 0.3 Wh. Capacitors with an energy storage capacity of 0.3 Wh or less are not subject to the provisions of the IMDG Code. Energy storage capacity means the energy held by a capacitor, as calculated using the nominal voltage and capacitance. All capacitors to which this entry applies, including capacitors containing an electrolyte that does not meet the classification criteria of any class or division of dangerous goods, shall meet the following conditions: 1. Capacitors not installed in equipment shall be transported in an uncharged state. Capacitors installed in equipment shall be transported either in an uncharged state or protected against short circuit; 2. Each capacitor shall be protected against a potential short circuit hazard in transport as follows: 1. when a capacitor's energy storage capacity is less than or equal to 10 Wh or when the energy storage capacity of each capacitor in a module is less than or equal to 10 Wh, the capacitor or module shall be protected against short circuit or be fitted with a metal strap connecting the terminals; and 2. when the energy storage capacity of a capacitor or a capacitor in a module is more than 10 Wh, the capacitor or module shall be fitted with a metal strap connecting the terminals; 3. Capacitors containing dangerous goods shall be designed to withstand a 95 kPa pressure differential; 4. Capacitors shall be designed and constructed to safely relieve pressure that may build up in use, through a vent or a weak point in the capacitor casing. Any liquid which is released upon venting shall be contained by the packaging or by the equipment in which a capacitor is installed; and 5. Capacitors manufactured after 31 December 2013 shall be marked with the energy storage capacity in Wh. Capacitors containing an electrolyte not meeting the classification criteria of any class or division of dangerous goods, including when installed in equipment, are not subject to other provisions of the IMDG Code. Capacitors containing an electrolyte meeting the classification criteria of any class or division of dangerous goods, with an energy storage capacity of 10 Wh or less are not subject to other provisions of the IMDG Code when they are capable of withstanding a 1.2 m drop test unpackaged on an unyielding surface without loss of contents. Capacitors containing an electrolyte meeting the classification criteria of				
Tank type:	Not allowed	Transport in bulk			Orange panel	
Tank provisions:	—	Not allowed			3499	
Stowage and segregation						
Stowage		There are no special requirements for stowage.				
Segregation		Check the segregation table for sea transport. There are no special requirements for segregation.				
Placarding of the transport unit		Containers, semi-trailers and tank containers must be marked with enlarged hazard labels (placards) on all four sides. Railway wagons must at least be marked with placards on both sides. All other transport units, e.g. trucks etc., must at least be placarded on both sides and at the back. The placards must have a size of at least 250 x 250 mm, and be of such quality that the labeling is still visible after 3 months at sea.				
Foodstuff		The substance is not required to be kept separate from food and feed.				
Limited quantities (LQ)				Excepted quantities (EQ)		
Combination packaging				Combination packaging		
Inner: Not allowed				Inner: Not allowed		
Outer:				Outer:		
Trays						
Inner: Not allowed						
Outer:						